

RANSOMWARE DEFENSE: BUILDING A BULLETPROOF BACKUP STRATEGY

MR SUNIL KUMAR



ABOUT ME



- Freelance corporate trainer
- Tech enthusiast
- Tech community builder
- YouTube: <https://www.youtube.com/@sunilkumarOnCloud>
- Medium blog: <https://medium.com/@suniel.vns>
- Topmate: <https://topmate.io/sunilkumar7>
- Twitter: @techwithsunil
- Toastmasters International member
- Runner

<https://www.linkedin.com/in/sunilkumardigi/>

AGENDA

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KNOW YOUR AUDIENCE QUIZ!

- [HTTPS://QUIZIZZ.COM/ADMIN/QUIZ/6262702594BAC3001D85BF66](https://quizizz.com/admin/quiz/6262702594BAC3001D85BF66)

RANSOMWARE

- RANSOMWARE IS A TYPE OF **MALWARE*** THAT INFECTS A COMPUTER AND RESTRICTS A USER'S ACCESS TO THE INFECTED SYSTEM OR SPECIFIC FILES IN ORDER TO EXTORT THEM FOR MONEY.

*SHORT FOR MALICIOUS SOFTWARE, IS ANY SOFTWARE INTENTIONALLY DESIGNED TO HARM, EXPLOIT, OR OTHERWISE COMPROMISE DEVICES, NETWORKS, OR DATA.

RANSOMWARE CONT..

- AFTER THE TARGET SYSTEM IS COMPROMISED, IT TYPICALLY LOCKS OUT MOST INTERACTION
- DISPLAYS AN ON-SCREEN ALERT, TYPICALLY STATING THAT THE SYSTEM IS LOCKED OR THAT ALL OF THEIR FILES HAVE BEEN ENCRYPTED.

EXAMPLE: WANNACRY RANSOMWARE- MAY 2017

ON-SCREEN ALERT- WANNACRY RANSOMWARE



Source: Wikipedia

MAP OF THE COUNTRIES INITIALLY AFFECTED BY WANNACRY RANSOMWARE



Source: Wikipedia

WHY ATTACKS SUCCEED

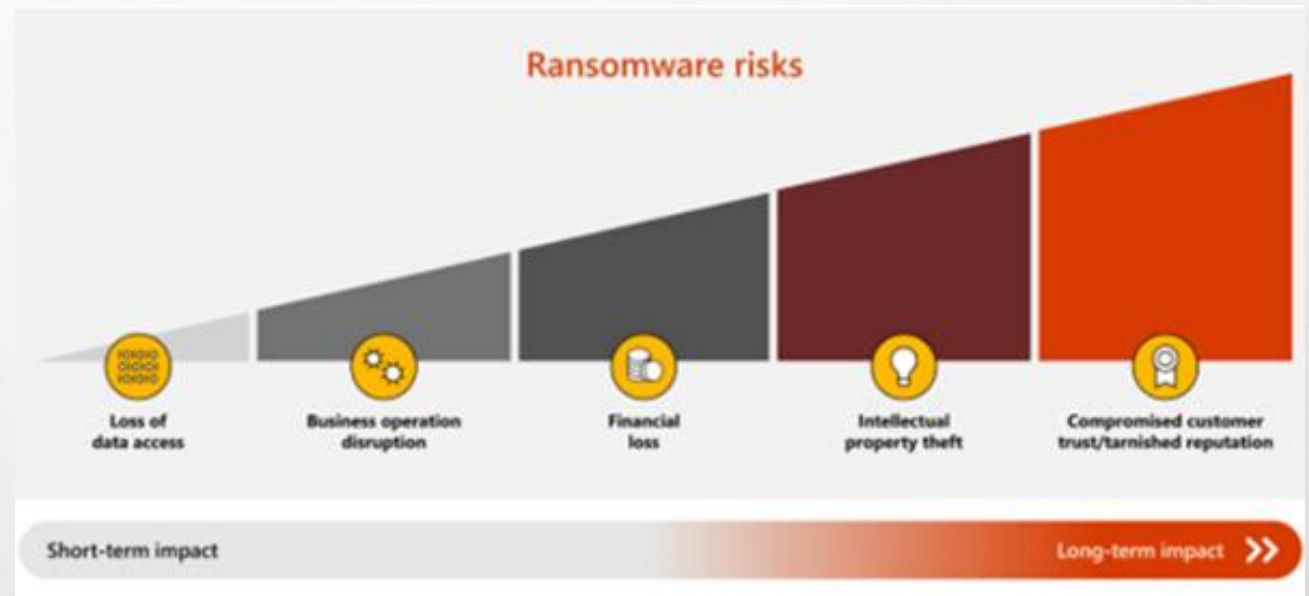
- **THE ATTACK SURFACE IS INCREASED** AS MORE BUSINESSES OFFER MORE SERVICES THROUGH DIGITAL OUTLETS.
- THERE'S A CONSIDERABLE EASE OF OBTAINING OFF-THE-SHELF MALWARE, **RANSOMWARE-AS-A-SERVICE (RAAS)**
- THE OPTION TO USE **CRYPTOCURRENCY FOR BLACKMAIL PAYMENTS** OPENS NEW AVENUES FOR EXPLOIT
- EXPANSION OF COMPUTERS AND THEIR USAGE IN DIFFERENT WORKPLACES EACH OF WHICH IS A POTENTIAL ACCESS POINT FOR MALWARE, RESULTING IN POTENTIAL ATTACK SURFACE
- **PREVALENCE OF OLD, OUTDATED INFRASTRUCTURE SYSTEMS AND SOFTWARE**

WHY ATTACKS SUCCEED CONT..

- POOR PATCH-MANAGEMENT REGIMENS
- OUTDATED OR OLD OPERATING SYSTEMS THAT ARE CLOSE TO OR HAVE GONE BEYOND END-OF-SUPPORT DATES
- LACK OF RESOURCES TO MODERNIZE THE IT FOOTPRINT
- KNOWLEDGE GAP
- LACK OF SKILLED STAFF AND KEY PERSONNEL OVERDEPENDENCY
- POOR SECURITY ARCHITECTURE

WHAT IS THE TYPICAL COST TO A BUSINESS?

- LOSS OF DATA ACCESS
- BUSINESS OPERATION DISRUPTION
- FINANCIAL LOSS
- INTELLECTUAL PROPERTY THEFT
- COMPROMISED CUSTOMER TRUST AND A TARNISHED REPUTATION



HOW CAN YOU PROTECT YOURSELF?

- THE BEST WAY TO PREVENT FALLING VICTIM TO RANSOMWARE IS TO **IMPLEMENT PREVENTIVE MEASURES**
- **PROTECT YOUR ORGANIZATION FROM EVERY STEP** THAT ATTACKERS TAKE TO INFILTRATE YOUR SYSTEMS.
- CSP'S HAVE INVESTED IN NATIVE SECURITY CAPABILITIES THAT MAKE THEM **RESILIENT** AGAINST RANSOMWARE ATTACKS AND HELPS ORGANIZATIONS DEFEAT RANSOMWARE ATTACK TECHNIQUES.

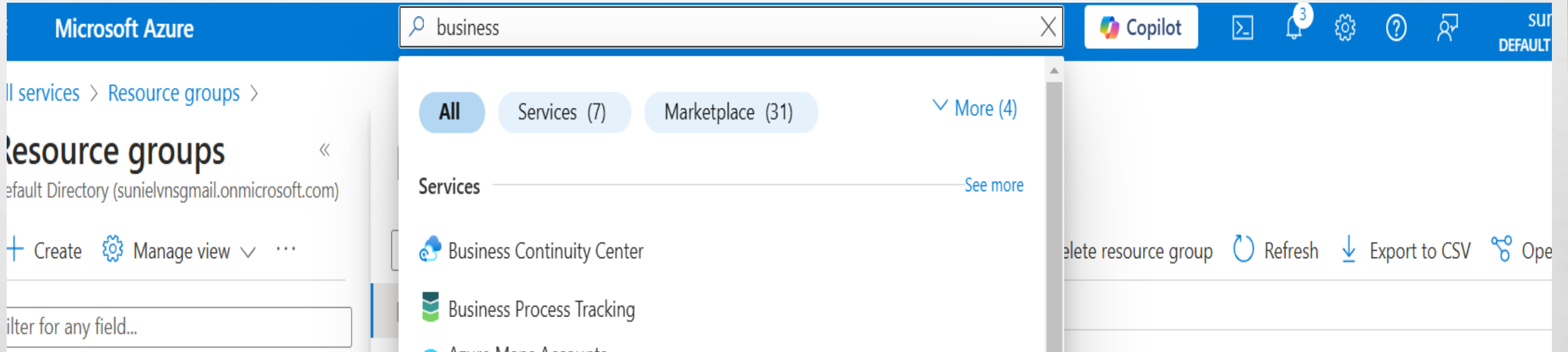
HOW CAN YOU PROTECT YOURSELF? CONT..

- ONE OF THE MOST IMPORTANT STEPS YOU CAN TAKE TO PROTECT YOUR DATA AND AVOID PAYING A RANSOM IS TO HAVE A **RELIABLE BACKUP AND RESTORE PLAN FOR YOUR BUSINESS-CRITICAL INFORMATION.**

AZURE BACKUP

- AZURE BACKUP PROVIDES SECURITY TO YOUR BACKUP ENVIRONMENT, BOTH WHEN YOUR DATA IS IN TRANSIT AND AT REST.
- WITH AZURE BACKUP, YOU CAN BACK UP:
 - ON-PREMISES FILES, FOLDERS, AND SYSTEM STATE
 - ENTIRE WINDOWS/LINUX VMS
 - AZURE MANAGED DISKS
 - AZURE FILE SHARES TO A STORAGE ACCOUNT
 - SQL SERVER DATABASES RUNNING ON AZURE VMS

AZURE BACKUP DEMO: BUSINESS CONTINUITY CENTER



AZURE BACKUP DEMO:-RECOVER SERVICE VAULT

The screenshot shows the Azure portal interface for the resource group 'thm_talk_demo'. The left sidebar contains navigation links for 'Resource groups', 'Overview', 'Activity log', 'Access control (IAM)', 'Tags', 'Resource visualizer', 'Events', 'Settings', 'Cost Management', 'Monitoring', 'Automation', and 'Help'. The main content area is titled 'thm_talk_demo Resource group' and includes a search bar and action buttons like 'Create', 'Manage view', 'Delete resource group', 'Refresh', 'Export to CSV', and 'Open query'. The 'Resources' tab is selected, showing a table of resources. The table has columns for 'Name', 'Type', and 'Location'. A single resource is listed: 'demo' with type 'Recovery Services vault' and location 'South India'. The 'Recovery Services vault' text is highlighted with a red box.

Name	Type	Location
demo	Recovery Services vault	South India

AZURE BACKUP DEMO: BACKUP GOAL

The screenshot shows the Microsoft Azure portal interface for a Recovery Services vault named "demo". The top navigation bar includes the Microsoft Azure logo, a search bar, and icons for Copilot, notifications (3), settings, help, and user profile. The left sidebar lists navigation options: Overview (selected), Activity log, Access control (IAM), Tags, Diagnose and solve problems, and Settings. The main content area features a search bar and a toolbar with icons for Backup, Enable Site Recovery, Delete, Refresh, and Feedback. A purple banner promotes the Business Continuity Center. Below this, the "Essentials" section is expanded, showing tabs for Overview (selected), Backup, and Site Recovery. The "What's new" section is also visible.

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

3

DEFA

All services >

demo Recovery Services vault

Search

+ Backup + Enable Site Recovery Delete Refresh Feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

> Settings

Try our new Business Continuity Center for the at scale BCDR management of your resources protected across Azure Backup and Site Recovery. →

Essentials

Overview Backup Site Recovery

What's new

AZURE BACKUP DEMO: BACKUP GOAL

 Microsoft Azure

Search resources, services, and docs (G+ /)

All services > demo >

Backup Goal ...

 The storage replication is set to Geo-Redundant. This option cannot be changed later. Before proceeding further, click here. →

Where is your workload running?

Azure

What do you want to backup?

 Virtual machine

Step: Configure Backup

Backup

Give feedback

 Help improve this page

AZURE BACKUP DEMO: CONFIGURE BACKUP

[All services](#) > [demo](#) > [Backup Goal](#) >

Configure backup

demo

Policy sub type *

☐ Enhanced

- ✓ Multiple backups per day
- ✓ Up to 30 days operational tier retention
- ✓ Support for Trusted Launch Azure VM
- ✓ Support for VMs with Ultra Disks and Premium SSD v2

☒ Standard

- ✓ Once-a-day backup
- ✓ Up to 5 days operational tier retention

Backup policy * ⓘ

DefaultPolicy

[Create a new policy](#)

i The list contains the policies pertaining to the selected policy sub type. [Learn more.](#)

Policy details

Full backup

Backup frequency
Daily at 5:00 PM UTC

Instant restore
Retain instant recovery snapshot(s) for 2 day(s)

Retention of daily backup point
Retain backup taken every day at 5:00 PM for 30 Day(s)

Consistency type ⓘ

Application or file-system consistent

Virtual machines

[Enable backup](#) [Download a template for automation](#)

AZURE BACKUP DEMO: ADD VIRTUAL MACHINE

All services > demo > Backup Goal >

Configure backup

demo

The list contains the policies pertaining to the selected policy and type. Learn more.

Policy details

Full backup

Backup frequency
Daily at 5:00 PM UTC

Instant restore
Retain instant recovery snapshot(s) for 2 day(s)

Retention of daily backup point
Retain backup taken every day at 5:00 PM for 30 days

Consistency type ⓘ
Application or file-system consistent

Virtual machines

NameResource group

No virtual machines selected.

Add

ⓘ

Selective disk backup option allows you to include or exclude specific data disks based on the backup type. This feature is not supported. Know more about Selective Disk Backup feature, its limitation and pricing.

Enable backup

Download a template for automation

Select virtual machines

Filter items by name

Virtual machine nameResource Group

There are no virtual machines that can be backed up in this vault.

< Previous

Page 1 of 1

Next >

OK

Cancel

VALIDATE BACKUPS

- VALIDATE THAT YOUR BACKUP IS GOOD AS YOUR BACKUP IS CREATED AND BEFORE YOU RESTORE.
- WE RECOMMEND THAT YOU USE A RECOVERY SERVICES VAULT, WHICH IS A STORAGE ENTITY IN AZURE THAT HOUSES DATA.

WHAT TO DO BEFORE AN ATTACK?

DETERMINE WHAT IS MOST IMPORTANT TO YOU

- **IDENTITY SYSTEMS** – REQUIRED FOR USERS TO ACCESS ANY SYSTEMS SUCH AS ACTIVE DIRECTORY, MICROSOFT ENTRA CONNECT, AD DOMAIN CONTROLLERS
- **HUMAN LIFE** – ANY SYSTEM THAT SUPPORTS HUMAN LIFE OR COULD PUT IT AT RISK SUCH AS MEDICAL OR LIFE SUPPORT SYSTEMS, SAFETY SYSTEMS LARGE MACHINERY, CHEMICAL/BIOLOGICAL SYSTEMS, PRODUCTION OF FOOD OR PERSONAL PRODUCTS, AND OTHERS
- **FINANCIAL SYSTEMS** – SYSTEMS THAT PROCESS MONETARY TRANSACTIONS AND KEEP THE BUSINESS OPERATING, SUCH AS PAYMENT SYSTEMS AND RELATED DATABASES, FINANCIAL SYSTEM FOR QUARTERLY REPORTING

WHAT TO DO BEFORE AN ATTACK? CONT..

- **PRODUCT OR SERVICE ENABLEMENT** – ANY SYSTEMS THAT ARE REQUIRED TO PROVIDE THE BUSINESS SERVICES OR PRODUCE/DELIVER PHYSICAL PRODUCTS THAT YOUR CUSTOMERS PAY YOU FOR, FACTORY CONTROL SYSTEMS, PRODUCT DELIVERY/DISPATCH SYSTEMS.
- **SECURITY (MINIMUM)** – PRIORITIZE THE SECURITY SYSTEMS REQUIRED TO MONITOR FOR ATTACKS AND PROVIDE MINIMUM SECURITY SERVICES. THIS SHOULD BE FOCUSED ON ENSURING THAT THE CURRENT ATTACKS (OR EASY OPPORTUNISTIC ONES) AREN'T IMMEDIATELY ABLE TO GAIN (OR REGAIN) ACCESS TO YOUR RESTORED SYSTEMS

STEPS TO TAKE BEFORE AN ATTACK

Identify the important systems that you need to bring back online first and immediately begin performing regular backups of those systems.

To get back up and running as quickly as possible after an attack, determine today what is most important to you.

Migrate your organization to the cloud. Consider purchasing a **Microsoft Unified Support** plan or working with a Microsoft partner to help support your move to the cloud.

Reduce your on-premises exposure by moving data to cloud services with **automatic backup** and self-service rollback. Microsoft Azure has a robust set of tools to help you back up your business-critical systems and restore your backups faster.

WHAT TO DO DURING AN ATTACK

Early in the attack, engage third-party support, particularly support from threat intelligence providers, anti-malware solution providers and from the malware analysis provider.

These contacts may be useful if the given ransomware variant has a known weakness or decryption tools are available.

The Microsoft Incident Response team can help protect you from attacks.

Contact your local or federal law enforcement agencies.

If you're in the United States, contact the FBI to report a ransomware breach using the IC3 Complaint Referral Form.

POST ATTACK

AFTER A RANSOMWARE ATTACK, TAKE THE FOLLOWING STEPS TO IMPROVE YOUR BACKUP AND RESTORE PLANS AS WELL AS YOUR SECURITY POSTURE:

- **IDENTIFY LESSONS** LEARNED WHERE THE PROCESS DIDN'T WORK WELL
- PERFORM **ROOT CAUSE ANALYSIS (RCA)** ON THE BIGGEST CHALLENGES
- INVESTIGATE AND REMEDIATE THE ORIGINAL BREACH (**ENGAGE THE MICROSOFT INCIDENT RESPONSE TEAM TO HELP**)
- **UPDATE YOUR BACKUP AND RESTORE STRATEGY** BASED ON LESSONS LEARNED AND OPPORTUNITIES — PRIORITIZING BASED ON HIGHEST IMPACT AND QUICKEST IMPLEMENTATION STEPS FIRST

REFERENCES

- [HTTPS://LEARN.MICROSOFT.COM/EN-US/AZURE/SECURITY/FUNDAMENTALS/BACKUP-PLAN-TO-PROTECT-AGAINST-RANSOMWARE](https://learn.microsoft.com/en-us/azure/security/fundamentals/backup-plan-to-protect-against-ransomware)
- WIKIPEDIA
- AZURE PORTAL

Q&A